

Key experience

Experience in prioritisation tools to inform policy on alien species:

- Helped establish a tool for the prioritisation of alien plants for biological control in South Africa – the Biological Control Target Selection System (BCTS) system. Application of the BCTS system to all alien plants listed in the South African National Environmental Management Biodiversity Act (NEM:BA) (<https://biocontrol-bcts.netlify.app/index.html>).
- Communicating with the Ministry of Forestry, Fisheries and the Environment, South Africa to provide guidance on future resource allocation for biological control targets based on the prioritised NEM:BA list.
- Performing horizon scan assessments on terrestrial invertebrates to prioritise species that present the greatest invasive threat to a region.

Invasive alien grass research:

- Completion of two grass identification courses run by the South African National Biodiversity Institute (SANBI).
- Working with the National Alien Grass Working group to improve collaboration on alien grass research and management in South Africa. Led communication with the Biosecurity Services Directorate to raise concerns on invasive alien grasses that pose a threat to South Africa.
- Reviewing alien grass risk assessments for ASRARP.
- Providing expert consultation to alien grass risk assessments.

Research on management of invasive alien plants in high elevation areas:

- Led research on understanding the current status of alien plants in South Africa's mountain areas.
- Member of the Southern African Mountains Invasive Alien Plant Working Group.
- Reviewing alien plant risk assessments for ASRARP for plants in high elevation areas.

Biological control of invasive alien plants:

- Surveying for natural enemies in their region of origin to identify potential biological control agents.
- Conducting genetic studies to determine the phylogeography of the target invasive alien plants.

Education

PhD Doctorate of Entomology

2017 Centre for Biological Control, Rhodes University, SOUTH AFRICA

BSc Bachelor of Science (Honours) - Environmental Monitoring and Modelling

2011 UNISA, SOUTH AFRICA

- Graduated with distinction

BSc Bachelor of Science - Environmental Science and Entomology

2007-2009 Rhodes University, SOUTH AFRICA

- Graduated with distinction in Environmental Science

Computer programs

- Geneious
- Microsoft suite
- Pastel accounting
- R
- Tableau

Languages

- English (fluent)
- French (basic)

IBD International Baccalaureate Diploma, International Community School
2002-2006 Addis Ababa, ETHIOPIA

Attended:

1999 - 2002 Dublin, Ireland
St. Gerard's School

1997 - 1999 Dublin, Ireland
Rathmichael National Parish School

1994 - 1997 Kampala, Uganda
Kabira International School

Work Experience and Skills

- 2025 – present Researcher, Centre for Biological Control, Rhodes University**
- Lead research projects for the Grassland Weeds programme
 - Responsible for writing grant proposals and communicating with funders
 - Implementing community outreach
 - Supervision of postgraduate students – currently 2 PhD, 3 MSc. students
- 2023 – present Research Fellow, Department of Plant Sciences and the Afromontane Research Unit, University of the Free State**
- Collaborate on invasive species research
 - Supervision of postgraduate students
- 2018 – July 2024 Post-Doctoral Research Fellow, Centre for Biological Control, Rhodes University**
- Helped establish a biological control prioritisation system for South Africa
 - Grass biocontrol specialist
 - Supervision of postgraduate students - 2 PhD, 2 MSc., 1 Honours graduated
- 2019 Canopy Growth, Lesotho**
- Consultant entomologist to train staff on pest identification and collection
- 2010-2020 Pure Imagination Media, South Africa and Lesotho**
- Company management including overseeing finances, human resources and client support.
- 2010-2011 Food and Trees for Africa, South Africa**
- Workshop Facilitator for the Eduplant programme to train educators on permaculture, pest management and sustainability.

Published Peer-reviewed Papers

Canavan K, Coetzee JA, Hill MP, Paterson ID (2014) Effect of water trophic level on the impact of the water hyacinth moth *Niphograpta albiguttalis* on *Eichhornia crassipes*. African Journal of Aquatic Science. 39(2): 203-208.

Canavan K, Paterson I, Hill MP (2014) The herbivorous arthropods associated with the invasive alien plant, *Arundo donax*, and the native analogous plant, *Phragmites australis*, in the Free State Province, South Africa. African Entomology. 1;22(2): 454-9.

- Visser V, Wilson JR, Canavan K, Canavan S, Fish L, Maitre DL, Nänni I, Mashau C, O'Connor TG, Ivey P, Kumschick S (2017) Grasses as invasive plants in South Africa revisited: Patterns, pathways and management. *Bothalia-African Biodiversity & Conservation*. 47(2):1-29.
- Canavan K, Paterson ID and Hill MP (2017) Exploring the origin and genetic diversity of the giant reed, *Arundo donax* in South Africa. *Invasive Plant Science and Management*, 10(1), pp.53-60.
- Canavan, S., Meyerson, L.A., Packer, J.G., Pyšek, P., Maurel, N., Lozano, V., Richardson, D.M., Brundu, G., Canavan, K., Cicatelli, A. and Čuda, J. (2018) Tall-statured grasses: a useful functional group for invasion science. *Biological Invasions*, 1-22.
- Canavan, K., Paterson, I.D., Hill, M.P. and Dudley T.L. (2018) Testing the Enemy release hypothesis on tall-statured grasses in South Africa, using *Arundo donax*, *Phragmites australis*, and *Phragmites mauritianus* as models. *Bulletin of Entomological Research*, 1-14, doi: 10.1017/S0007485318000627.
- Canavan, K., Paterson, I.D., Lambertini, C. and Hill M.P (2018) Expansive reed populations – alien invasion or disturbed wetlands? *AoB Plants*, doi: 10.1093/aobpla/ply014.
- Sutton, G.F., Canavan, K., Day, M.D., den Breeyen, A., Goolsby, J.A., Cristofaro, M., McConnachie, A. and Paterson, I.D (2019) Grasses as suitable targets for classical weed biological control. *BioControl*, 1-18, doi: 10.1007/s10526-019-09968-8.
- Canavan, K., Canavan, S., Harms, N.E., Lambertini, C., Paterson, I.D. and Thum, R. (2020) The potential for biological control on cryptic plant invasions. *Biological Control*, p.104243.
- Mapaura, A., Canavan, K., Richardson, D.M., Clark, V.R. and Steenhuisen, S.L. (2020) The invasive grass genus *Nassella* in South Africa: A synthesis. *South African Journal of Botany*, 135, 336-348.
- Sutton, G.F., Canavan, K., Day, M.D. and Paterson, I.D. (2021) Field-based ecological studies to assess prospective biological control agents for invasive alien plants: An example from giant rat's tail grass. *Journal of Applied Ecology*, 58: 1043-1054.
- Canavan K, Paterson ID, Ivey P, Sutton GF, Hill MP (2021) Prioritisation of targets for weed biological control III: A tool to identify the next targets for biological control in South Africa and set priorities for resource allocation. *Biocontrol Science and Technology*. 31: 584-601.
- Paterson ID, Hill MP, Canavan K, Downey PO (2021) Prioritisation of targets for weed biological control II: The South African Biological Control Target Selection system. *Biocontrol Science and Technology*. 31: 566-583.
- Downey PO, Paterson ID, Canavan K, Hill MP (2021) Prioritisation of targets for weed biological control I: A review of existing prioritisation schemes and development of a system for South Africa. *Biocontrol Science and Technology*. 31: 546-565.
- Canavan K, Canavan S, Clark VR, Gwate O, Richardson DM, Sutton G, Martin G (2021) The alien plants that threaten South Africa's mountain ecosystems. *Land*. 10(12): 1393.
- Sutton G, Visser V, Bownes A, Canavan K (2021) Progress and prospects for the biological control of invasive grasses in South Africa. *African Entomology*. 29(3): 837-858.
- Canavan K, Magengelele NL, Paterson ID, Williams DA, Martin GD (2022) Uncovering the phylogeography of *Schinus terebinthifolia* in South Africa to guide biological control. *AoB Plants*. 14(1): plab078.
- Canavan K., Canavan S., Clark VR., Gwate O., Mapaura A., Richardson DM., Steenhuisen S., Martin G. (2022) Alien plant invasions in South Africa's mountains: impacts and management options. In: *Exploring the multiple values of nature - connecting ecosystems and people across landscapes*. Springer 30: 169.
- Verlaque R, Hardion L, Lambertini C, Canavan K, Verlaque M, Vila B. (2022) New highlights on Old World giant *Phragmites* (Poaceae) using leaf and floral bract microscopic characters. *Aquatic Botany*. Oct 17:103591.

- Novoa A., Canavan K., Canavan S. 2022. Ch. 9. The role of second homes in non-native plant invasions. In: *Tourism, Recreation and Biological Invasions*, Barros A., Shackleton R., Rew L.J., Pizarro C., Pauchard A. (eds.), CABI, Gloucester, UK.
- Goolsby JA, Moran PJ, Jiménez MM, Yang C, Canavan K, Paynter Q, Ota N, Kriticos DJ 2023. Biology of Invasive Plants 4. *Arundo donax* L. Invasive Plant Science and Management.:1-29.
- Canavan K, Paterson ID 2023. Prioritization of alien plant targets for biological control in South Africa's offshore sub-Antarctic islands. *Antarctic Science*.:1-8.
- Gwate O, Canavan K, Martin GD, Richardson DM, Ralph Clark V 2023. Assessing habitat suitability for selected woody range-expanding plant species in African mountains under climate change. *Transactions of the Royal Society of South Africa*. 17:1-5.
- Sutton, G.F., van Steenderen, C.J.M., Yell, L.D., Canavan, K., McConnachie, A. and Paterson, I.D., 2023. First record of an African grass-feeding wasp (Tetramesa; Eurytomidae) on the invasive grass *Eragrostis curvula* (African lovegrass; Poaceae) in Australia. *BiolInvasions Records*, 12(3):673-680.
- Sutton, G.F., Day, M.D., Canavan, K., and Paterson, I.D., 2023. Anthropogenic disturbance affects specialist, but not generalist, endophagous insects associated with two African grasses: implications for biological control. *Biocontrol Science and Technology*, doi.org/10.1080/09583157.2023.2275114.
- Lieurance, D., Canavan, S., Behringer, D.C., Kendig, A.E., Minter, C.R., Reisinger, L.S., Romagosa, C.M., Flory, S.L., Lockwood, J.L., Canavan K., Anderson, P.J. and Baker, S.M., 2023. Identifying invasive species threats, pathways, and impacts to improve biosecurity. *Ecosphere*, 14(12), p.e4711.
- Yell, L.D., Sutton, G.F., van Steenderen, C.J.M., Canavan, K., McConnachie, A., Paterson, I.D., 2024. Field-based surveys and laboratory tests indicate that candidate biocontrol agents for African lovegrass from South Africa are not suitable for release in Australia. *BioControl Science and Technology*, doi: 10.1080/09583157.2024.2317135.
- Lázaro-Lobo, A., Andrade, B.O., Canavan, K., Ervin, G.N., Essl, F., Fernández-Pascual, E., Follak, S., Richardson, D.M., Moles, A., Visser, V. and Wyse, S.V., 2024. Monographs on invasive plants in Europe N° 8: *Cortaderia selloana* (Schult. & Schult. f.) Asch. & Graebn. *Botany Letters*, pp.1-25.
- Mapaura, A., Canavan, K., Richardson, D.M., Clark, V.R., Sutton G.F. and Steenhuisen, S.L. 2024. The impact of *Nassella trichotoma* (Nees) Hack. ex Arechav. on plant diversity, richness and soil properties in South Africa. *South African Journal of Botany*, 173, 175-183.
- Martin, G., Canavan, K., Chikowore, G., Bugan, R., De Lange, W., du Toit, B., Harding, G., Heath, R., Hill, M., Hurley, B.P. and Ivey, P., 2025. Managing wilding pines in the Cape Floristic Region, South Africa: Progress and prospects. *South African Journal of Botany*, 177, pp.377-391.
- Smith, M.C., Canavan, K., Minter, C.R., Lieurance D. 2025. Preemptive and proactive application of biological control for weeds: An argument for swifter action to aid conservation efforts. *Biological Control*, 202, 105725.
- Bertheau, C., Roques, A., Hoffmann, J., Canavan, K., Martin, G. and Roux, G., 2025. Exploring the taxonomic status of the Palearctic cone weevil, *Pissodes validirostris*, inferred from morphometric and molecular data. *BioControl*, pp.1-13.
- van Steenderen, C.J., Sutton, G.F., Yell, L.D., Canavan, K. and Paterson, I.D., 2025. Species delimitation and molecular dating of southern African Tetramesa (Hymenoptera: Eurytomidae): An understudied microhymenopteran group. *Austral Entomology*, 64(2), p.e70006.
- Canavan, S., Canavan, K., Kumschick, S., Gordon, D.R., Wilson, J.R. and Lieurance, D., 2025. Invasive species risk assessment in practice: Insights from a survey of practitioners. *NeoBiota*, 99, pp.341-362.
- Mbele, T.J., Steenhuisen, S.L., Payne, S.L. and Canavan, K., 2025. Seed germination of cultivated, naturalised and floristically traded *Cortaderia* species in South Africa. *South African Journal of Botany*, 186, pp.152-162.
- Martin, G.D., Canavan, K., Westwood, T., Chikowore, G., Fraser, G., Clark, Q., Payne, S. and Steenhuisen, S.L., 2025. Prickly politics: How *Rosa rubiginosa* (Sweet-Briar Rose) divides rural communities, private landowners, and commercial players in South Africa. *South African Journal of Botany*, 186, pp.701-708.

- Mapaura, A., Canavan, K., Richardson, D.M., Clark, V.R., Junior J.D.V and Steenhuisen, S.L. 2025. Implications of climate change for the distribution of the invasive grass genus *Nassella* in South Africa and Lesotho. *Rangeland Ecology and Management*, 103, 367-376.
- Canavan, S., Pipek, P., Canavan, K., Jarić, I., Healy, K., Lieurance, D., Pattison, Z., Pyšek, P., Stevenson, E.A. and Novoa, A., 2025. From habitats to hashtags: examining online discussions about invasive species. *Ecology and Society*, 30(4).
- Mapaura, A., Canavan, K., Richardson, D.M., Le Maitre, N.C., Ralph Clark, V. and Steenhuisen, S.L., 2025. DNA-based species identification as a tool to distinguish invasive *Nassella* species in South Africa. *African Journal of Range & Forage Science*, 42(4), pp.339-342.
- Dube, K.D., Chikowore, G., Canavan, K., Martin, G.D., and Paterson I., 2026. Landscape-scale patterns, temporal shifts, and impacts of invasive alien plants species in a montane grassland of Eastern Zimbabwe. *Transactions of the Royal Society of South Africa*, <https://doi.org/10.1080/0035919X.2026.2672397>.

Selected Conference and Workshop Presentations

- Canavan K., Paterson I.D, Hill M.P. *Arundo donax* in South Africa. 2013. Ecology and Management of Alien Plant Invasions (EMAPI), Oral Presentation: The potential for biological control of the Giant Reed, *Arundo donax* in South Africa. Paper No. SS5.5. Pirenópolis, Brazil.
- Canavan K., Paterson I.D. 2013. Determining the origin of the common reed, *Phragmites australis* in South Africa base on chloroplast DNA. Proceedings of the 41st Annual Symposium on Management of Invasive Alien Plants, Cape St. Francis, South Africa.
- Canavan K., Paterson I.D, Hill M.P. 2013. Herbivores associated with the alien invasive Giant Reed, *Arundo donax* compared to its native analogous species, *Phragmites australis*, in South Africa. Proceedings of the XVIII Congress of the Entomological Society of Southern Africa, Potchefstroom.
- Canavan, K., Canavan, S., Harms, N.E., Lambertini, C., Paterson, I.D. and Thum, R. 2019. Cryptic invasions and weed biological control. National Symposium on Biological Invasion. Tulbagh, 15 -17 May.
- Canavan, K., Paterson, I.D., Lambertini, C. and Hill M.P. 2020. *Phragmites australis* – alien invasion or disturbed wetlands? South African Association of Botanists, Qwaqwa, 7-10 January.
- Canavan, K., Paterson, I.D., Ivey, P., Sutton, G.F., Bownes, A., Visser, V., Mapaura, A. & Hill, M.P. 2022. The South African Biological Control Target Selection System. *National Symposium on Biological Invasions*. 6-8 July 2022. University of Fort Hare, Alice, Eastern Cape.
- Canavan, K., Paterson, I.D., Ivey, P., Sutton, G.F., Bownes, A., Visser, V., Mapaura, A. & Hill, M.P. 2022. The South African Biological Control Target Selection System. *The 48th Weeds Workshop*. 18-20th October 2022. Golden Gate Highlands National Park.
- Canavan, K., Paterson, I.D., Ivey, P., Sutton, G.F., & Hill, M.P. 2023. The South African Biological Control Target Selection System. *XVI International Symposium on Biological Control of Weeds*. 7th- 12th May 2023. Iguazú, Argentina.
- Canavan, K., Canavan, S., Clark, V.R., Gwate, O., Mapaura, A., Richardson, D.M., Steenhuisen, S.-L., Sutton, G., Martin, G.D. 2023. Invasive alien plants in South Africa's mountains. *National Symposium on Biological Invasions*. 4-6 July 2023. Houw Hoek Hotel, Western Cape.
- Canavan, K., Canavan, S., Ralph Clark, V., Gwate O., Mapaura, A., Richardson D.M., Steenhuisen S., Sutton G.F, Martin G.D. 2024. Elevation patterns of non-native invasions in South African mountains. *The 59th Annual Congress of the Grassland Society of Southern Africa*. 22-26 July 2024. Gariep Forever Resort, Free State.
- Canavan, K., Martin G.D. 2024. Managing wilding Pines in the Cape Floristic Region, South Africa: progress and prospects. *National Symposium on Biological Invasions*. 9-12 September 2024. Sol Plaatje University, Kimberley, Northern Cape.

Canavan, K., Chikowore, G., Hill, M., Martin, G. Managing wilding pines in South Africa's Cape Floristic Region. XVII International Symposium on Biological Control of Weeds, Rotorua, Aotearoa, New Zealand, 8-13 March 2026.

Canavan K., Invasive Alien Plants in Africa. Leadership for Conservation in Africa. Launch of 'Unlocking Nature' series. Online event, 18th June 2026.

Scientific review

Journals and organisations reviewed for:

The Alien Species Risk Analysis Review Panel, African Entomology, Annals of Applied Biology, Aquatic Ecology, Biological Control, Biocontrol Science and Technology, Biological Invasions, Environmental Management, Frontiers in Ecology and Evolution, National Research Foundation, Peer J, Plant Ecology and Evolution, Regional Environmental Change

Research Activities

Scopus H-Index: 14

Research Interest Score: 538.7

Number of journal articles: 41

Total number of citations (August): 585

Professional affiliations and public engagement

- | | |
|-----------------------|---|
| 2005-2006 | Vice President of the National Honours Society (NHS)
Voted in as vice president of the NHS to help guide students to achieve scholarship, leadership and service. |
| 2008 | Dean's Merit List
Placed on the Dean's Merit list in the faculty of Science for academic achievement. |
| 2011 | Founded the Bophelo Food Garden Project
The food garden project was established in the rural township of Marquard. Clinic staff completed a training course on how to maintain food gardens after which they could assist and encourage the community to set up their own gardens. Resources have also been distributed to ensure access to planting materials. |
| 2022 – 2024 | Environmental education, Ichthus Academy
Established an environmental club to help children learn about environmental issues and how to contribute towards conservation efforts. |
| 2013 – present | Member of the National Working Group on Alien Grasses
Work with the group to improve collaboration and research on alien grasses in South Africa. This has included helping to run meetings, co-authoring publications and providing reviews for alien grass risk assessments. |
| 2018 – present | Board Member, Ichthus Academy – private school, Marquard, South Africa |

Serving on the management board to help guide the academic programme, school financing and administration.

2018- present Chair of the Board at the Mama Sophia centre for individuals living with disabilities, Marquard, South Africa

Elected in as chair of the management board of this organisation. Responsibilities includes assisting with obtaining government financial subsidies, guiding the centre's programme and administration.

2021 – present Member of the Southern African Mountains Invasive Alien Plant Working Group

Work with the group to improve collaboration among researchers and conservation managers for improved management and research on invasive alien plants in southern African mountains. This has involved leading a publication that reviewed the status of alien invasive plants in South Africa's mountains.

Collaborations

Local collaboration :

- South African National Biodiversity Institute (SANBI)
- Centre for Invasion Biology (CIB)

International collaboration:

- Agronomy Department, University of Florida (Florida, USA)
- Department of Botany, Texas Christian University (Texas, USA)
- United States Department of Agriculture (USA)
- University of California (California, USA)
- Marseille herbarium (Marseille, France)
- University of Bologna (Bologna, Italy)
- Academy of Sciences of the Czech Republic (Prague, Czech Republic)
- University of Alcalá (Alcalá de Henares, Spain)
- Penn State University (Pennsylvania, USA)

Referees

1. Dr. Sandy-Lynn Steenhuisen, Afromontane Research Unit and Department of Plant Sciences, University of the Free State. steenhuisens@ufs.ac.za.
2. Professor David Mark Richardson, Centre for Invasion Biology. rich@sun.ac.za